

# CHRISTOPHER FU

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## EDUCATION

### University of California, Los Angeles (UCLA)

M.S. Data Science in Biomedicine

Graduation Date: Jun. 2026

Cumulative GPA: 4.00/4.00

**Relevant Coursework:** Foundations of DS (Probability and Statistics), ML Applications in Biomedicine, DS for Medical Imaging, Research

**Awards:** University of California, Los Angeles Warren Alpert Computational Biology and Artificial Intelligence Scholar, Dean's Honor List

### University of California, Los Angeles (UCLA)

B.S. Data Theory

Graduation Date: Jun. 2024

Cumulative GPA: 3.79/4.00

## PROFESSIONAL EXPERIENCE

### Warren Alpert Computational Biology and AI Network

Los Angeles, CA

*Warren Alpert Computational Biology and AI Network Fellow*

Sep. 2025 - Dec. 2025

- Built an end-to-end pipeline combining longitudinal post-mortem MRI data, SynthSeg, and autopsy reports to quantify brain structure volumes and deformation rates across patients, linking imaging-based biomarkers with ground truth labels for neurodegeneration analysis
- Developed scalable analysis workflows including 3D segmentation validation, patient-level heatmaps of anatomical volumes and deformation trends, and a pair plot of autopsy variables to assess feature relationships, producing a dataset of 1300 rows for our cohort
- Applied LOOCV logistic regression to evaluate performance of brain structures in Alzheimer's Disease classification, achieving 65% accuracy with high F1 on select features while underscoring sparse, interpretable biomarkers and limitations of small, imbalanced cohorts

### Kaiser Permanente

Pasadena, CA

*Applied Data and Analytics Master's Intern*

Jun. 2025 - Sep. 2025

- Performed ETL (EDA, feature engineering) in Python on 1K physician survey responses, identifying demographic and event-level drivers of engagement; partnered with the CME research consultants to improve survey design and integrate findings into a future AI Summit
- Fine-tuned Hugging Face transformer models (Few-Shot learning) on 950K dermatology patient messages, collaborating with a dermatologist to build a message classifier for triage (Macro F1 = 0.891) ready for Fall 2025 deployment to improve clinician workflows
- Adapted Kaiser Permanente's primary care message classification model across 5 medical specialties, identifying low-volume underperforming classes; provided recommendations to enhance specialty-level classification accuracy for predicting patient messages

### UCLA Biodesign

Los Angeles, CA

*Biodesign AI Fellow*

Sep. 2024 - Jun. 2025

- Employed SQL to query, transform, and clean OB/GYN patient flow healthcare data from electronic health records (EHR) of 80,956 rows spanning 2022–2024; discovered bed usage, delivery, & departmental efficiency trends by exploratory analysis (line plot, bar plot)
- Designed, implemented, and evaluated an LSTM model to forecast OB capacity time-series trends, training on 2022–2023 data; achieved a test RMSE of 17.69, providing actionable insights to optimize resource and staff allocation, as well as enhance operational planning
- Partnered with a team of six to analyze patient flow data within a school's comprehensive data repository; developed a project strategy, presented findings, and proposed actionable next steps, emphasizing opportunities to improve efficiency and data-driven decisions

### Children's Hospital of Orange County (CHOC)

Orange, CA

*Data Science Intern*

Jun. 2023 - Aug. 2023

- Analyzed demographic disparities among diagnosed anxiety patients by constructing multiple linear regression models using Python's Statsmodels and SciPy libraries, achieving 77% accuracy; conducted t-tests and other statistical analyses to identify actionable insights
- Led a team of four data science interns in delivering data-driven insights to physicians through visually engaging presentations; created bar plots, box plots, model summary tables, and geographical heat maps to highlight trends and promote predictive measures in mental health
- Engineered and proposed detailed action plans for deploying advanced unsupervised machine learning techniques to uncover latent factors contributing to anxiety disorders in pediatric patients, supporting data-driven interventions and improved outcomes

## SKILLS

**Programming:** Python (NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, Jupyter, Tensorflow, Keras), SQL (Azure Data Studio), R (ggplot2)

**Data Analysis & Machine Learning:** Regression, Classification, Unsupervised (Clustering), Neural Networks (LSTM, CNN), NLP

**BI & Visualization Tools:** Tableau, PowerBI, MS Excel, Matplotlib, Seaborn, MS Powerpoint, MS Word, MS Office, Dashboards